

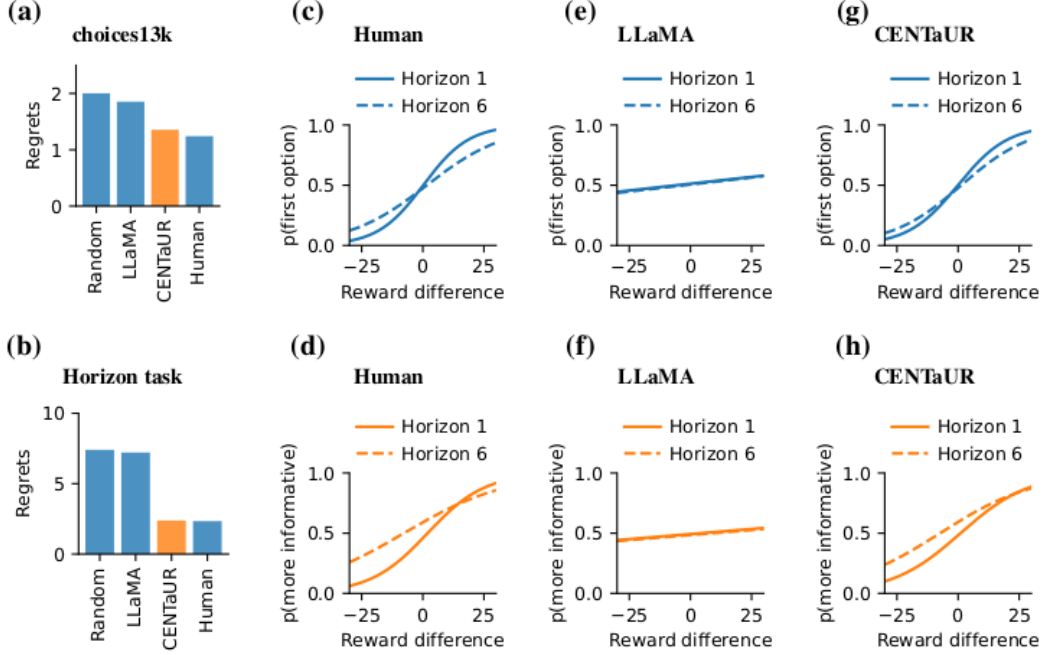


Figure 2: Model simulations. (a) Performance for different models and human participants on the choices13k data set. (b) Performance for different models and human participants on the horizon task. (c) Human choice curves in the equal information condition of the horizon task. (d) Human choice curves in the unequal information condition of the horizon task. (e) LLaMA choice curves in the equal information condition of the horizon task. (f) LLaMA choice curves in the unequal information condition of the horizon task. (g) CENTaUR choice curves in the equal information condition of the horizon task. (h) CENTaUR choice curves in the unequal information condition of the horizon task.

3 Model simulations reveal human-like behavior

We next verified that CENTaUR shows human-like behavioral characteristics. To do so, we simulated the model on the experimental data. Looking at performance, we found that finetuning led to models that closely resemble human performance as shown in Figure 2a and b. For the choices-13k data set, CENTaUR obtained a regret (defined as the difference between the highest possible reward and the reward for the action selected by the model) of 1.35 (SE = 0.01), which was much closer to the human regret ($M = 1.24$, SE = 0.01) than the regret of LLaMA ($M = 1.85$, SE = 0.01). The results for the horizon task showed an identical pattern, with CENTaUR ($M = 2.38$, SE = 0.01) matching human regret ($M = 2.33$, SE = 0.05) more closely than LLaMA ($M = 7.21$, SE = 0.02).

In addition to looking at performance, we also inspected choice curves. For this analysis, we took the data from the first free-choice trial in the horizon task and divided it into two conditions: (1) an equal information condition that includes trials where the decision-maker had access to an equal number of observations for both options and (2) an unequal information condition that includes trials where the decision-maker previously observed one option fewer times than the other. We then fitted a separate logistic regression model for each condition with reward difference, horizon, and their interaction as independent variables onto the simulated choices. Earlier studies with human subjects [Wilson et al., 2014] identified the following two main results regarding their exploratory behavior: (1) people’s choices become more random with a longer horizon in the equal information condition (as shown in Figure 2c) and (2) people in the unequal information condition select the more informative option more frequently when the task horizon is longer (as shown in Figure 2d). While LLaMA did not show any of the two effects (see Figure 2e and f), CENTaUR exhibited both of them (see Figure 2g and h), thereby further corroborating that it accurately captures human behavior.

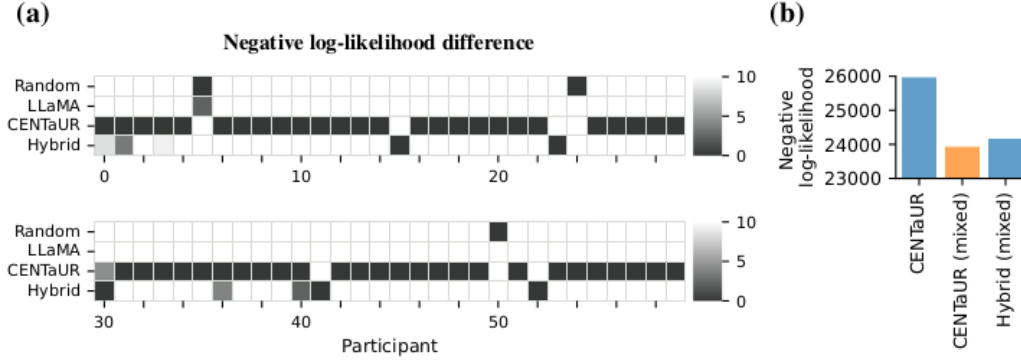


Figure 3: Individual differences. (a) Negative log-likelihood difference to the best-fitting model for each participant. Black highlights the best-fitting model, while white corresponds to a difference larger than ten. (b) Negative log-likelihoods for models that were finetuned using the random-effects structure described in the main text.

4 Language model embeddings capture individual differences

We also investigated how well CENTaUR describes the behavior of each individual participant. Note that this form of analysis is only possible for the horizon task as choice information on the participant level is not available for the choices13k data set. In total, the majority of participants ($N = 52$ out of 60) was best modeled by CENTaUR (see Figure 3a for a detailed visualization). We furthermore entered the negative log-likelihoods into a random-effects model selection procedure which estimates the probability that a particular model is the most frequent explanation within a set of candidate models [Rigoux et al., 2014]. This procedure favored CENTaUR decisively, assigning a probability that it is the most frequent explanation of close to one.

Thus far, we have finetuned LLaMA jointly for all participants. However, people may exhibit individual differences that are not captured by this analysis. To close this gap and test whether LLaMA embeddings can account for individual differences, we incorporated random effects in the finetuned layer. We added a random effect for each participant and embedding dimension while keeping the remaining evaluation procedure the same. Figure 3b illustrates the resulting negative log-likelihoods. Including the random-effect structure improved goodness-of-fit considerably ($NLL = 23929.5$) compared to the fixed-effect-only model ($NLL = 25968.6$). Furthermore, CENTaUR remained superior to the hybrid model with an identical random-effect structure ($NLL = 24166.0$). Taken together, the findings reported in this section highlight that embeddings of large language models contain the information necessary to model behavior on the participant level.

5 Evaluating goodness-of-fit on hold-out tasks

Finally, we examined whether CENTaUR – after being finetuned on multiple tasks – is able to predict human behavior in an entirely different task. This evaluation protocol provides a much stronger test for the generalization abilities of our approach. Following our initial analyses, we finetuned a linear layer on top of LLaMA embeddings. However, this time, we fitted a joint model using both the data from the choices13k data set and the horizon task, and then evaluated how well the finetuned model captures human choices on a third task. Further details about the fitting procedure are provided in the Materials and Methods section. For the hold-out task, we considered data from a recent study that provided participants with a choice between one option whose information is provided via a description and another option for which information is provided via a list of experienced outcomes [Garcia et al., 2023]. Figure 4a shows an example prompt for this experimental paradigm.

Finetuning was generally beneficial for modeling human behavior on the hold-out task: negative log-likelihoods for CENTaUR ($NLL = 4521.1$) decreased both in comparison to a random guessing model ($NLL = 5977.7$) and LLaMA ($NLL = 6307.9$). We were thus curious whether CENTaUR also captures human behavior on a qualitative level. To test this, we took a look at the key insight from the original study: people tend to overvalue options that are provided through a description

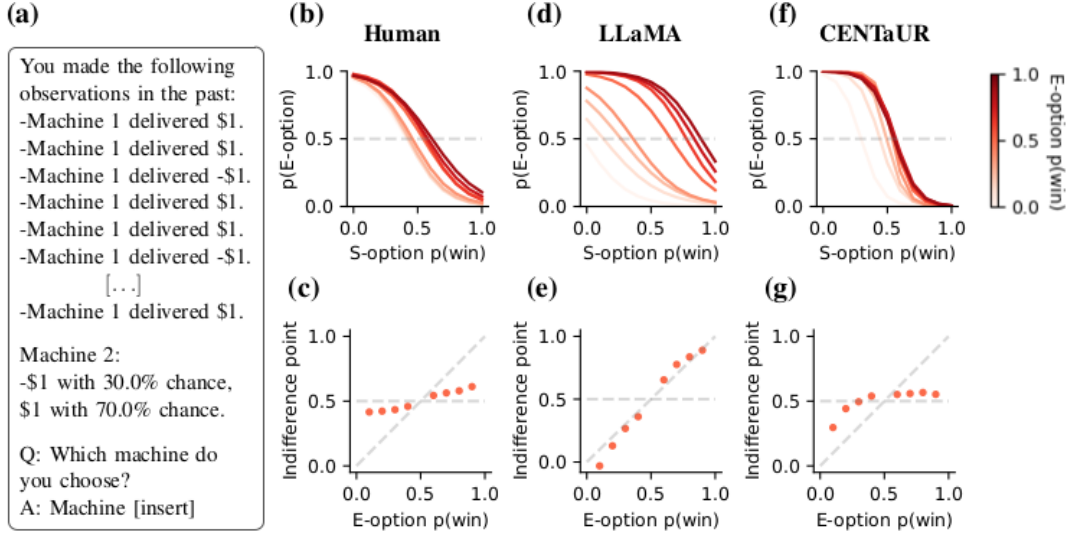


Figure 4: Hold-out task evaluations. (a) Example prompt for the experiential-symbolic task of Garcia et al. [2023]. (b) Human choice curves as a function of win probabilities for both options. (c) Human indifference points as a function of win probability for the E-option. Indifferent points express the win probabilities at which a decision-maker is equally likely to select both options. (d) LLaMA choice curves as a function of win probabilities for both options. (e) LLaMA indifference points as a function of win probability for the E-option. (f) CENTaUR choice curves as a function of win probabilities for both options. (g) CENTaUR indifference points as a function of win probability for the E-option.

(symbolic or S-options) over the options that come with a list of experienced outcomes (experiential or E-options) as illustrated in Figure 4b and c. LLaMA does not show this characteristic and instead weighs both option types equally (Figure 4d and e). In contrast to this, CENTaUR shows human-like behavior, taking mostly the S-option into account (Figure 4f and g). This is remarkable because we never presented data from the experiment under consideration during finetuning.

6 Discussion

We have demonstrated that large language models can be turned into cognitive models by finetuning their final layer. This process led to models that achieved state-of-the-art performance in two domains. Furthermore, these models were able to capture behavioral differences at the individual participant level. Finally, we have shown that our approach generalizes to previously unseen tasks. In particular, a model that was finetuned on two tasks also exhibited human-like behavior on a third, hold-out task.

These results complement earlier work showing that large language model embeddings allow us to predict behavior and neural activations in linguistic settings [Schrimpf et al., 2021, Kumar et al., 2022, Tuckute et al., 2023, Antonello et al., 2023]. For example, Schrimpf et al. [2021] showed that large language models can predict neural and behavioral responses in tasks that involved reading short passages with an accuracy that was close to noise ceiling. While it may be expected that large language models explain human behavior in linguistic domains (after all these models are trained to predict future word occurrences), the observation that these results also transfer to more cognition domains like the ones studied here is highly non-trivial.

We are particularly excited about one feature of CENTaUR: embeddings extracted for different tasks all lie in a common space. This property allows finetuned large language models to solve multiple tasks in a unified architecture. We have presented preliminary results in this direction, showing that a model finetuned on two tasks can predict human behavior on a third. However, we believe that our current results only hint at the potential of this approach. Ideally, we would like to scale up our approach to finetuning on a larger number of tasks from the psychology literature. If one would include enough tasks in the training set, the resulting system should – in principle – generalize to *any*